Abhijit Karale

Embedded Systems & Firmware Engineer | STM32 | ARM Cortex-M | FreeRTOS | Hardware Bring-Up
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PROFESSIONAL SUMMARY

Embedded Systems and Firmware Engineer with 2+ years of production experience in firmware architecture and board bring-up. Expertise in STM32 / ARM Cortex-M, Nordic nRF52 BLE, FreeRTOS preemptive multitasking, and peripheral driver development. Adept in laboratory hardware debugging: multi-layer PCB fault isolation, oscilloscope signal analysis, and analog conditioning.

TECHNICAL SKILLS

- Microcontrollers & Platforms: STM32, ARM Cortex-M0/M3/M4, Nordic nRF52840/nRF52833, LPC1768, 8/32-bit MCUs
- Firmware & Software: Embedded C, C++, FreeRTOS (tasks, queues, mutexes, semaphores), Bare-metal, Bare-metal HAL, Git
- Communication Protocols: UART, SPI, I2C, BLE 5.0 (Low-Power System ON/OFF), RS-485, Modbus, ADC, PWM, GPIO
- Hardware & Lab Tools: Multi-layer PCB Debugging, Signal Integrity, Oscilloscopes, Logic Analyzers, Multimeters, Soldering
- Development Environments: Keil uVision, STM32CubeIDE, Nordic nRF Connect SDK, J-Link, OpenOCD

PROFESSIONAL EXPERIENCE

Velaxtra Pvt Ltd â **Sr Embedded Firmware Engineer** Oct 2025 - Jun 2026 | Bangalore

- Architected and implemented production embedded C firmware for smart industrial and remote-asset monitoring devices.
- Developed non-blocking drivers for UART, SPI, I2C, ADC, and PWM peripherals managed under FreeRTOS priority queues.
- Accelerated board bring-up by correlating firmware execution traces with hardware oscilloscope captures.

Minilec India Pvt Ltd â **Embedded System Engineer** Jul 2024 - Sep 2025 | Pune

- Engineered STM32 and ARM Cortex-M control firmware for high-volume industrial motor protection and monitoring relays.
- Diagnosed analog signal-conditioning circuits, resolved PCB signal-integrity issues, and eliminated intermittent field faults.
- Standardized factory production testing and hardware verification procedures for electronic relay units.

Saksham Semiconductors â **Analog Circuit Design Intern** May 2024 - Jul 2024 | Hyderabad

- Simulated MOSFET characteristics and designed differential amplifiers and current mirrors in SPICE.

The Silicon Sandbox â **Trainee, Advanced RTL & Verification** Jun 2026 - Present | Bangalore

- Gained advanced RTL architecture, SystemVerilog, and UVM verification training bridging firmware with silicon design.

KEY HARDWARE & EMBEDDED PROJECTS

Multi-Peripheral Industrial Firmware Engine STM32, FreeRTOS, Embedded C

- Built robust preemptive multitasking framework integrating 5 peripherals (UART, SPI, I2C, ADC, PWM) with zero task starvation.

Ultra-Low-Power BLE Firmware for Edge Devices Nordic nRF52840, BLE, Low Power

- Optimized battery autonomy via fine-grained System ON / System OFF sleep states and RTC wake-up triggers.

Industrial SCR Motor Drive Firing Package Diagnostics Komatsu/Joy SCR 601849-121

- Isolated firing-circuit failure modes in high-power industrial SCR motor drive modules and established testing protocol.

EDUCATION & LEADERSHIP

MAEER's MIT College of Railway Engineering & Research, Barshi â **B.Tech | EXTC** 2021 - 2025 | CGPA: 7.86
MAEER's MIT College of Railway Engineering & Research â **Diploma in Railway Engineering** 2018 - 2021 | 82.26%
Vice President â **ETSA (Electronics & Telecomm Students Association)** 2024 - 2025